

DNIPROPETROVSK, Ukraine

WMO: 345040

Lat: 48.3600N

Lon: 35.0850E

Elev: 143

StdP: 99.62

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-14.8	-20.8	0.6	-17.5	-17.7	0.8	-14.3	12.6	-11.5	11.6	-9.9	4.2	30	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	34.1	20.9	32.1	20.4	30.4	19.9	22.6	30.4	21.7	29.2	20.9	28.1	4.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	15.1	25.8	19.2	14.2	24.6	18.4	13.5	23.7	67.3	30.4	63.9	29.4	61.0	28.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.5	9.3	8.3	DB	-20.6	36.9	3.9	2.3	-23.5	38.5	-25.8	39.9	-28.0	41.2	-30.8	42.9
(5)				WB	-20.9	24.7	3.9	1.4	-23.7	25.7	-26.0	26.4	-28.1	27.2	-31.0	28.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.9	-2.6	2.6	10.6	16.9	20.7	23.0	22.4	16.6	9.5	3.2	-1.8	(6)
(7)		DBStd	10.66	5.56	5.48	4.41	4.20	4.35	3.62	3.26	3.77	4.31	4.64	4.65	5.21	(7)
(8)		HDD10.0	1705	431	353	230	43	3	0	0	3	67	209	365		(8)
(9)		HDD18.3	3540	689	586	487	234	80	17	3	7	82	277	455	623	(9)
(10)		CDD10.0	1646	0	0	2	60	217	322	402	386	202	50	5	0	(10)
(11)		CDD18.3	437	0	0	0	1	36	89	146	134	30	2	0	0	(11)
(12)		CDH23.3	4172	0	0	0	18	354	809	1375	1331	272	13	0	0	(12)
(13)		CDH26.7	1647	0	0	0	3	105	287	573	597	81	2	0	0	(13)
(14)	Wind	WSAvg	4.1	4.4	4.6	4.7	4.2	3.7	3.8	3.6	3.9	3.9	4.2	4.5	(14)	
(15)	Precipitation	PrecAvg	509	41	37	36	38	44	59	52	47	35	36	44	49	(15)
(16)		PrecMax	852	135	77	90	90	140	154	130	216	119	127	125	126	(16)
(17)		PrecMin	333	4	2	4	0	0	7	1	2	2	2	0	4	(17)
(18)		PrecStd	118	29	18	21	22	29	33	35	40	29	29	29	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	10.0	17.2	25.6	32.1	33.9	36.5	37.8	32.1	25.1	16.2	10.2	(19)
(20)		MCWB	6.9	7.8	9.7	14.7	19.0	20.8	22.1	20.9	18.7	16.2	12.6	9.1	(20)	
(21)		2%	DB	5.7	7.1	14.0	22.3	29.3	31.8	33.9	35.0	28.9	21.3	13.7	8.2	(21)
(22)		MCWB	5.1	5.7	8.5	13.1	18.2	20.3	21.5	20.5	17.5	14.9	11.5	7.5	(22)	
(23)		5%	DB	4.0	5.1	11.2	20.1	27.2	30.0	32.0	32.2	26.3	18.8	11.4	6.3	(23)
(24)		MCWB	3.6	4.1	7.1	12.4	17.6	19.9	20.9	20.1	16.8	13.8	10.0	5.7	(24)	
(25)		10%	DB	2.2	3.5	9.0	18.1	24.9	28.1	30.0	30.0	24.0	16.0	9.4	4.9	(25)
(26)		MCWB	1.7	2.7	6.1	11.1	16.6	19.3	20.5	19.6	16.0	12.7	8.4	4.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	8.5	11.0	15.7	20.9	23.1	24.2	23.2	20.6	17.8	13.6	9.5	(27)
(28)		MCDWB	7.4	9.4	15.8	23.5	28.0	30.2	31.4	31.8	28.2	22.4	15.5	10.1	(28)	
(29)		2%	WB	5.0	5.9	9.2	14.1	19.4	21.7	23.0	22.1	19.0	15.9	11.9	7.6	(29)
(30)		MCDWB	5.3	6.8	12.9	20.5	26.8	28.8	30.8	30.6	25.7	19.3	13.5	8.2	(30)	
(31)		5%	WB	3.4	4.1	7.6	13.0	18.3	20.8	22.0	21.1	17.8	14.4	10.3	5.8	(31)
(32)		MCDWB	3.9	4.9	10.5	18.8	25.3	27.8	29.5	29.4	23.8	17.9	11.3	6.2	(32)	
(33)		10%	WB	1.8	2.7	6.2	12.0	17.2	19.8	21.2	20.2	16.8	13.0	8.5	4.4	(33)
(34)		MCDWB	2.1	3.3	8.5	17.0	24.0	26.4	28.1	27.9	22.3	15.8	9.3	4.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	5.1	6.1	7.8	10.4	11.4	11.0	11.4	11.9	11.0	8.8	5.8	4.7	(35)	
(36)		5% DB	MCDBR	5.0	6.9	11.9	13.6	14.1	13.4	13.8	14.8	14.1	12.3	8.2	5.5	(36)
(37)		MCWBR	4.5	5.6	7.5	6.8	6.4	5.6	5.3	5.2	6.3	6.9	6.0	5.0	(37)	
(38)		5% WB	MCDBR	4.7	6.4	10.8	12.2	12.6	11.8	12.1	12.7	12.1	10.8	7.5	5.4	(38)
(39)		MCWBR	4.4	5.4	7.3	6.4	6.2	5.6	5.4	5.3	6.0	6.8	5.9	5.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.356	0.386	0.449	0.415	0.412	0.437	0.436	0.409	0.370	0.352	0.331	(40)	
(41)		taud	2.214	2.166	2.211	2.082	2.270	2.298	2.233	2.227	2.269	2.358	2.315	2.274	(41)	
(42)		Ebn at Noon	708	775	818	798	849	852	822	804	786	760	681	655	(42)	
(43)		Edh at Noon	80	105	119	150	129	126	133	128	112	87	73	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.93	1.81	2.97	4.51	5.84	6.24	6.19	5.57	3.85	2.30	1.03	0.67	(44)	
(45)		RadStd	0.18	0.35	0.39	0.41	0.49	0.53	0.38	0.35	0.47	0.34	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.79	N/A	N/A	+1.12	N/A	N/A	-175	-218	+117	+76
(47)	Regional (9 neighbors)	+0.78	N/A	N/A	+1.08	N/A	N/A	-144	-206	+126	+80

Nomenclature: See separate page